Training Leaves Traces: Weight-Level Model Provenance Without Watermarks

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Abstract

As foundation models are increasingly fine-tuned, merged, and redistributed, verifying whether a checkpoint derives from a protected model is critical for IP protection and compliance—yet current methods require hashes, metadata, or watermarks that fail when checkpoints are adapted or records are incomplete. We ask whether *weight-level lineage* can be verified from weights alone: given a reference and suspect checkpoint of compatible architecture, did the suspect inherit the reference's weights?

We find that trained residual architectures expose such a signal. Training couples a residual block's input and output projections so that their product exhibits diagonal dominance absent from mismatched or random pairings. We formalize this with the score $s(M) = |\mathrm{tr}(M)|/\|M\|_F$, derive a separation margin scaling as $\Theta(\sqrt{d})$, and build a two-level pipeline: diagonal dominance recovers block correspondences; centered residual signatures detect checkpoint lineage.

Across 12 architectures (GPT-2, BERT, Mistral, LLaMA-2, Qwen, DeepSeek, Whisper, ResNet, ViT, ConvNeXt) spanning 5 families and 280+ pairs, the fingerprint recovers 91–100% of block correspondences, and the lineage score achieves AUROC=1.000 (TPR=100% at 1% FPR). The signal persists under fine-tuning, quantization, pruning, and RLHF; suppressing it below the detection threshold incurs 12–100% eval-loss degradation. The method detects weight inheritance, not functional imitation: distilled students correctly classify as non-descendants because they share no weights.

1 Introduction

The open-weight ecosystem has grown explosively: Hugging Face hosts nearly one million model repositories, with derivatives proliferating faster than originals (Hugging Face 2024). When Meta's LLaMA weights leaked, the community produced instruction-tuned variants—Alpaca, Vicuna, Koala—within weeks (Touvron et al. 2023). This rapid adaptation means any widely-distributed checkpoint spawns hundreds of fine-tuned, merged, and quantized descendants whose lineage quickly becomes untraceable.

Current provenance methods have fundamental limitations. Cryptographic hashes become invalid after any weight modification. Metadata and logs (Mitchell et al. 2019) can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) require insertion before deployment, and no surveyed scheme proved robust against fine-tuning, pruning, or extraction attacks (Lukas et al. 2022). Decision-boundary fingerprints (Cao, Jia, and Gong 2021) detect functional similarity but cannot distinguish weight lineage from distillation. These limitations leave a gap: no method provides robust weight-level lineage verification for already-deployed checkpoints.

We ask: *does training itself leave a recoverable signature in the relational structure between weight matrices?* The answer is yes. In residual architectures, the product of a block's input and output projections develops diagonal dominance—trace/Frobenius concentration absent from mismatched or random pairings. This emerges because residual blocks must approximate identity mappings for stable gradient flow, a constraint formalized as *dynamical isometry* (Pennington, Schoenholz, and Ganguli 2017). We formalize the signal with $s(M) = |\mathrm{tr}(M)|/\|M\|_F$ and derive a separation margin scaling as $\Theta(\sqrt{d})$.

This enables *passive weight-level lineage verification*: given reference and suspect checkpoints, decide whether the suspect inherited the reference's weights—without watermarks, metadata, or cooperation. The fingerprint emerges from residual optimization and can be read retroactively, surviving fine-tuning, quantization, and pruning while degrading only when perturbations cause $>12\%$ utility loss.

Two tasks, one mechanism. *Block correspondence* asks which input projection pairs with which output projection within a model; the diagonal-dominance score suffices. *Checkpoint lineage* asks whether one model descends from another; this requires centered residual signatures and is reported as AUROC. Both share the same mechanism but have different failure modes. The method applies to residual architectures and detects weight inheritance, not functional imitation—distilled students correctly classify as non-descendants.

Across 12 architectures (GPT-2, BERT, LLaMA-2, Mistral, ResNet, ViT, Whisper) spanning 5 families and 280+ checkpoint pairs, block correspondence achieves 91–100% accuracy, and lineage detection achieves AUROC=1.000 with TPR=100% at 1% FPR. No prior method addresses passive weight-level lineage for post-hoc checkpoints; existing watermarks require insertion before training and fail under fine-tuning (Lukas et al. 2022), while decision-boundary fingerprints confuse distillation with weight inher-

Method	Retroactive	Survives FT	Survives Quant	Distill-aware
Crypto Hash	✓	✗	✗	✓
Metadata/Logs	✓	✓	✓	✓
Watermarking	✗	∼	∼	✗
Decision-Boundary FP	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retroactive: works on already-deployed models. Distill-aware: correctly rejects distilled copies as non-descendants. ∼: partial/unreliable.

itance (Cao, Jia, and Gong 2021).

We contribute: (1) formalization of diagonal dominance with a $\Theta(\sqrt{d})$ margin analysis verified on GPT-2; (2) a passive verification protocol using Hungarian matching and centered signatures; (3) validation across 12 architectures with robustness under fine-tuning, quantization, pruning, and RLHF.

2 Background and Problem Setting

We define *passive model provenance*: given a reference checkpoint R and suspect S with missing or untrusted metadata, determine whether S shares weight-level lineage with R after post-training transformations. This is a white-box task—the verifier accesses both weight sets. The method detects *weight-level lineage* (parameters derived via fine-tuning, pruning, or quantization), not behavioral equivalence. Distilled students and model-extraction copies produce similar outputs but share no weights; our method correctly classifies them as non-descendants.

Table 1 positions our approach against prior methods. Cryptographic hashes fail after any modification. Metadata can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) require insertion before deployment and lack robustness (Lukas et al. 2022). Decision-boundary fingerprints (Cao, Jia, and Gong 2021) confuse distillation with weight inheritance. Our key differentiator: we read a signal that training *leaves*, rather than one that must be *inserted*.

Modern residual networks (He et al. 2016) compute $x' = x + F(x)$, where F is the residual branch. For stable gradient flow, the Jacobian $J = I + J_F$ must remain near-orthogonal—*dynamical isometry* (Pennington, Schoenholz, and Ganguli 2017). For a branch with K linear layers, we define the *residual branch product*:

$$M = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \tag{1}$$

Table 2 shows the architecture-aware factorization. Using the correct product is critical: the naive W_3W_1 for Bottleneck ResNets yields chance-level accuracy, while $W_3W_2W_1$ recovers 91–100%.

Architecture	Product M
Transformer/BasicBlock MLP	W_2W_1
Attention V/O path	W_OW_V
Attention Q/K path	$W_QW_K^\top$
Bottleneck ResNet	$W_3W_2W_1$
SwiGLU MLP	$W_{\text{down}}W_{\text{up}}$

Table 2: Architecture-aware factorization for residual branch products.

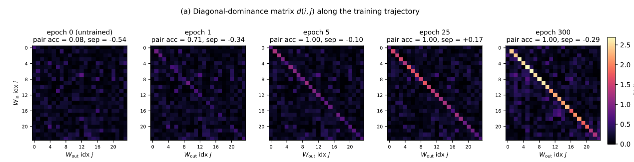

Figure 1: Diagonal dominance emergence. Left: At initialization, no diagonal structure. Right: By epoch 5, 100% pair accuracy. The fingerprint is learned, not an initialization artifact.

3 Training-Induced Diagonal Dominance

Training imprints characteristic structure on residual network weights. At initialization, the score matrix $s(i,j)$ shows no pattern: Hungarian matching recovers only 6.25% of correct pairs in a 48-block network. Critically, orthogonal initialization provides dynamical isometry by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs—ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. By epoch 5, the diagonal is fully separated with 100% pair accuracy; the signal emerges rapidly, well before loss plateaus (Figure 1).

For a branch product $M_{ij} = W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}$, we define the *diagonal-dominance score*:

$$s(i,j) = \frac{|\text{tr}(M_{ij})|}{\|M_{ij}\|_F} \tag{2}$$

Correctly paired blocks score $\Theta(\sqrt{d})$; mismatched pairs score $\Theta(1/\sqrt{d})$. The gap grows with dimension d, making wider networks more separable.

Dynamical isometry (Pennington, Schoenholz, and Ganguli 2017) predicts that trained residual blocks have Jacobian $J = I + W_{\text{out}}W_{\text{in}}$ close to orthogonal. We decompose:

$$M = W_{\text{out}}^{(i)} W_{\text{in}}^{(i)} = -\varepsilon I + E \tag{3}$$

where $\varepsilon = |\text{tr}(M)|/d$ and $\text{tr}(E) = 0$. The following proposition quantifies the margin.

Proposition 1 (Margin Formula)**.** *Let $M = -\varepsilon I + E$ with $\text{tr}(E) = 0$. Then:*

$$s(i,i) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \tag{4}$$

with equality iff $M = -\varepsilon I$.

Proof. We have $|\text{tr}(M)| = \varepsilon d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ (using $\text{tr}(E) = 0$). Substituting: $s(i,i) = \varepsilon d/\sqrt{\varepsilon^2 d + \|E\|_F^2} = \sqrt{d}/\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}$. □

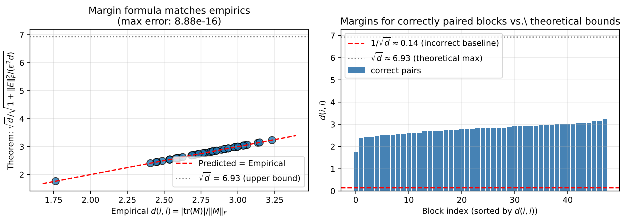

Figure 2: Proposition 1 verification on GPT-2. Left: Predicted vs. empirical $s(i,i)$. Right: Correct-pair margins vs. baseline and $\sqrt{d}$ bound.

Corollary 1 (Signal-to-Baseline)**.** *For mismatched pairs with independent random entries, $\mathbb{E}[s(i,j)] \approx 1/\sqrt{d}$. The signal-to-baseline ratio scales as d.*

On GPT-2 models, mean $s(i,i)$ rises from 4.18 (d=768) to 7.77 (d=1600), following the $\sqrt{d}$ envelope. Figure 2 confirms the margin formula reproduces empirical scores to floating-point precision.

The fingerprint requires both training *and* residual connections. A PlainNet (no skip connections) achieves only 3% pair accuracy vs. 100% for ResNet, despite similar loss (Appendix A). The skip connection creates the functional constraint—blocks must approximate identity—that imprints diagonal structure

4 From Fingerprints to Provenance Verification

4.1 Block Pairing via Hungarian Matching

Given reference R and suspect S with L blocks each, we construct a score matrix $\mathbf{S} \in \mathbb{R}^{L \times L}$:

$$s(i,j) = \frac{|\mathrm{tr}(M_{ij})|}{\|M_{ij}\|_F}, \quad M_{ij} = \tilde{W}_{\mathrm{out}}^{(j)} W_{\mathrm{in}}^{(i)} \tag{5}$$

If S descends from R, corresponding blocks yield $s(i,j) = \Theta(\sqrt{d})$; non-corresponding pairs remain at $\Theta(1/\sqrt{d})$. We recover correspondences via Hungarian matching (Kuhn 1955):

$$\pi^* = \arg\max_{\pi \in \mathcal{P}_L} \sum_{i=1}^{L} s(i, \pi(i)) \tag{6}$$

On GPT-2 (L=12), this achieves 100% pair accuracy with 35× separation ratio.

4.2 Model-Level Lineage Score

Block pairing establishes correspondences; provenance requires a scalar statistic. For each branch ℓ, we compute a *centered residual signature* by subtracting the generic diagonal component:

$$R_\ell = M_\ell - \tfrac{\mathrm{tr}(M_\ell)}{d} I, \quad \phi_\ell = \tfrac{\mathrm{vec}(R_\ell)}{\|\mathrm{vec}(R_\ell)\|_2} \tag{7}$$

This isolates checkpoint-specific structure: R_ℓ is exactly the E_ℓ term from $M_\ell \approx -\varepsilon I + E_\ell$. The lineage score aggregates

Suspect Type	n	Mean $\mathcal{L}$	Range
Fine-tune / Quantization / Noise	45	0.99	[0.97, 1.00]
Fine-tune (different task)	15	0.94	[0.84, 1.00]
Magnitude pruning (10–85%)	15	0.81	[0.58, 1.00]
Independent / Random init	75	0.08	[0.01, 0.20]
Distilled student	9	0.09	[0.03, 0.19]

Table 3: Lineage score distributions. Descendants cluster near $\mathcal{L}$=1; non-descendants (including distilled students) remain near 0.08. Separation ≈ 0.72.

Transformation	Parameters	Detection
Fine-tuning	LR 10^{-5}–10^{-3}	100%
Quantization	8-bit / 6-bit	100% / 94%
Pruning	30% / 50%	100% / 98%
LoRA merge	rank 8–64	100%

Table 4: Verification robustness at 95% confidence threshold.

aligned branch similarities:

$$\mathcal{L}(A,B) = \frac{1}{L} \sum_{\ell=1}^{L} \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \tag{8}$$

This satisfies self-similarity ($\mathcal{L}(A,A) = 1$), symmetry, and boundedness in $[-1, 1]$.

Table 3 shows score distributions on depth-24 MLPs. Descendants achieve $\mathcal{L} \geq 0.58$ even under 85% pruning; non-descendants stay below 0.20. This yields AUROC=1.000 and TPR=100% at 1% FPR. A baseline using only $s(M)$ achieves AUROC=0.417 (chance), confirming that centering isolates lineage-specific information.

4.3 Verification Protocol

The protocol outputs DESCENDANT, NON-DESCENDANT, or INCONCLUSIVE based on a z-score against a null distribution of unrelated models. Table 4 summarizes robustness: 100% detection under fine-tuning, 8-bit quantization, 30% pruning, and LoRA merging; degradation only when perturbations destroy utility. Full protocol details in Appendix C.

5 Experiments

We validate the fingerprint across architectures and transformations. Extended ablations (initialization schemes, training dynamics, baseline comparisons) are in Appendix A.

5.1 Cross-Architecture Generalization

Table 5 reports block-pairing accuracy across 12 architectures spanning language (GPT-2, BERT, LLaMA-2, Mistral, Qwen, DeepSeek), vision (ViT, ConvNeXt, ResNet), and speech (Whisper). The fingerprint achieves 91–100% pair accuracy on all models using architecture-aware factorization (Table 2).

Key findings: (1) The fingerprint transfers across GELU, SwiGLU, causal/bidirectional attention, and grouped-query

Model	Layers	Pair Acc	AUC
GPT-2 (124M–1.5B)	12–48	**100%**	1.00
BERT-base / ViT-B/16	12	**100%**	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	**100%**	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	**100%**	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	4/5→joint	0.93–1.00
Whisper (tiny/base/small)	4–12	**100%**	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 5: Cross-architecture generalization. Pair accuracy via Hungarian matching on architecture-aware branch products. Sub-100% SwiGLU sub-paths are rescued by joint factorization. Random-init baselines: $\leq 9\%$.

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	—

Table 6: Lineage score survival. All 75 descendants detected at 1% FPR; non-descendants stay below 0.20.

attention. (2) RLHF and reasoning distillation preserve the signal (LLaMA-2 base→chat: 100% accuracy). (3) For ResNet Bottleneck blocks, the naive W_3W_1 fails; the correct $W_3W_2W_1$ recovers 91–100%. (4) GPT-2 scores follow the $\sqrt{d}$ envelope: s=4.18 at d=768 to s=7.77 at d=1600.

5.2 Robustness Under Post-Training Modification

We test survival under five transformation families on depth-24 MLPs: fine-tuning (same/different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), and quantization (16–256 levels).

Table 6 shows 100% detection at 1% FPR across all transformations. The weakest descendant (85% pruning, $\mathcal{L}$=0.58) remains 7× above the strongest non-descendant (distilled, $\mathcal{L}$=0.19). Fingerprint degradation correlates with utility loss ($\rho = -0.83$): suppressing the signal damages the model.

Real architectures: On CIFAR-10 ResNet-18, the method achieves AUROC=1.000 with >100× separation between descendants and independents (Appendix A).

6 Robustness and Boundary Cases

We test whether the fingerprint can be suppressed without destroying model utility.

Function-preserving symmetries. Per-block permutations ($W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^{\top}$) and cross-layer rescaling ($W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$) leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged, so $\mathcal{L} = 1.0$ under these attacks. Orthogonal rotations also preserve M but break model function.

Gradient-based suppression. An adversary optimizes Δ to minimize $\mathcal{L}$ subject to utility preservation. Sweeping the utility weight λ traces a Pareto frontier: with $\lambda \geq 1$, the

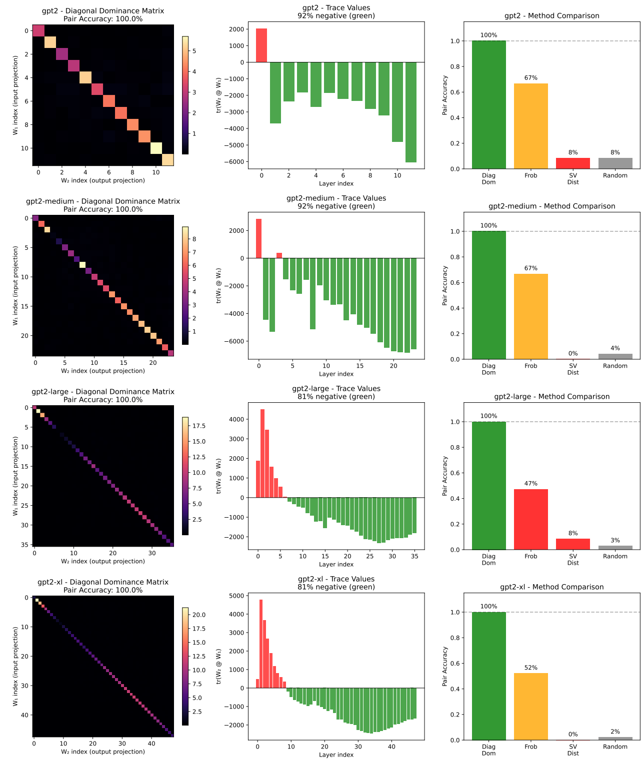

Figure 3: GPT-2 pairing. Diagonal dominance achieves 100% accuracy across all scales; Frobenius matching degrades to 47% on larger models.

attacker cannot push $\mathcal{L}$ below 0.91; suppressing to chance level ($\mathcal{L} \approx 0.08$) requires ≥12% eval-loss degradation; full suppression destroys the model (57× loss increase). Details in Appendix E.

Distillation boundary. The fingerprint tracks weight inheritance, not functional imitation. Distilled students ($\mathcal{L} = 0.086$) are correctly classified as non-descendants—they trained fresh weights and share no residual structure with the teacher, even though they replicate its outputs.

Cross-layer routing architectures. A natural concern is whether the fingerprint depends on the standard $x + f(x)$ identity path. We test this directly against DAR (Xu et al. 2026), a recent residual replacement that drops the additive recurrence in favor of a learnable, timestep-adaptive softmax aggregation over sublayer history. At matched fingerprint plateau on depth-24 MLPs, the negative-identity component is essentially ablated under DAR (negative-trace fraction $0.99 \rightarrow 0.02$), confirming that this specific geometry is regime-dependent. However, the broader trace/Frobenius concentration in the architecture-aware product persists: separation between correct and mismatched pairs is comparable (0.42 vs. 0.39 for standard residual), and the centered residual-signature score yields AUROC=1.000 across noise, prune, quant, and fine-tune descendants (45 desc / 36 indep pairs, Figure 5). The lineage signal is therefore not tied to the negative-identity form of dynamical isometry; it follows from trace concentration in the trained branch product more

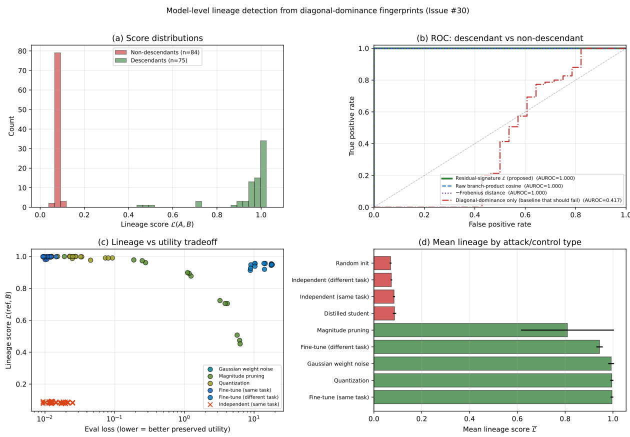

Figure 4: Lineage detection on depth-24 residual MLPs. (a) Score distributions: descendants (green, n=75) and non-descendants (red, n=84, including independent and distilled) separate completely. (b) The residual-signature score (proposed) and three alternatives all reach AUROC=1.000, except the diagonal-dominance-only baseline (AUROC=0.417) which cannot distinguish two trained residual models. (c) Lineage vs. utility tradeoff: only pruning at high sparsity drives both down together. (d) Mean lineage by control type: descendants $\geq$ 0.81, non-descendants ≤ 0.09.

generally.

7 Discussion and Limitations

Scope. The method requires white-box access and residual architecture. Non-residual networks lack the identity-target constraint that creates the fingerprint.

Weight-level, not behavioral. The signature is a property of weights, not function. Distilled students ($\mathcal{L} \approx 0.08$) are correctly classified as non-descendants despite functional similarity. The method cannot detect training-data overlap or knowledge transfer.

Attacks. Function-preserving symmetries (permutations, rescaling) leave $\mathcal{L} = 1$ exactly—these are invariances, not vulnerabilities. Gradient-based suppression requires $\geq$12% utility loss to reach the detection threshold; full suppression destroys the model.

Positioning. This complements watermarking: watermarks require forethought; diagonal dominance offers passive, retroactive verification for checkpoints never watermarked.

8 Conclusion

Training leaves traces. Residual-network optimization imprints a diagonal-dominance fingerprint in branch products ($W_{\text{out}} W_{\text{in}} \approx -\varepsilon I + E$), with separation margin scaling as $\Theta(\sqrt{d})$. Across 12 architectures spanning 5 families, the fingerprint recovers 91–100% of block correspondences and achieves AUROC=1.000 for lineage detection, surviving fine-tuning, quantization, pruning, and RLHF. Suppressing the fingerprint below the detection threshold requires $\geq$12% utility loss. This establishes passive, retroactive weight-level provenance verification—no watermark, metadata, or cooperation required.

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A Extended Experimental Details

A.1 Initialization Ablation

We test whether the diagonal-dominance fingerprint depends on initialization scheme. We train depth-24 resid-

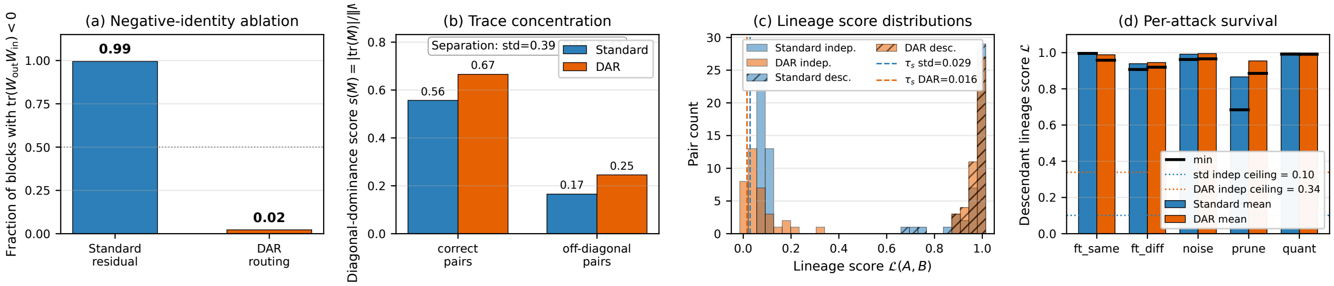

Figure 5: Fingerprint survives DAR cross-layer routing (Xu et al. 2026) on depth-24 MLPs at matched plateau. (a) Negative-identity component is ablated under DAR ($0.99 \rightarrow 0.02$). (b) Trace concentration on correct pairs is preserved; separation from off-diagonal pairs is comparable to standard residual. (c) Lineage score distributions remain bimodal with AUROC=1.000 for both architectures. (d) Per-attack descendant scores stay above each architecture's independent ceiling across noise, pruning, quantization, and same/different-target fine-tuning.

ual MLPs (h=64, d=24) on a synthetic regression target for 200 epochs from seven initialization schemes: Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal. Table 7 reports pair accuracy before and after training.

Init scheme	Untrained	Trained	AUC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian σ=0.02	2.1%	13%	0.671

Table 7: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq$2.1%) at initialization, rising to 75–100% after training. Orthogonal initialization—which provides dynamical isometry at init—shows *zero* correct pairs before training.

Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. The σ=0.02 small-Gaussian case fails because residual blocks collapse to near-zero contribution; without active per-block contribution there is no Jacobian constraint to enforce.

A.2 PlainNet Control Experiment

To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical to ResNet except that skip connections are removed.

While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline—with AUC 0.51 and no discernible diagonal structure. The skip connection creates a functional constraint that imprints the characteristic trace structure.

Architecture	Eval Loss	Pair Acc	AUC	Neg Trace
ResNet-24	1.35	100%	0.98	100%
PlainNet-24	1.56	3%	0.51	47%

Table 8: ResNet vs. PlainNet control. Despite similar evaluation loss, PlainNet achieves only chance-level pair accuracy (3% vs. chance $\approx 4.2\%$), confirming the skip connection is necessary for the fingerprint.

A.3 Training Dynamics

The fingerprint emerges rapidly during training. On a 48-block residual network:

- Epoch 0: Mean correct-pair score = 0.19, pair accuracy = 6.25%
- Epoch 5: Diagonal fully separated, pair accuracy = 100%
- Epoch 300: Mean correct-pair score = 2.07, mean incorrect-pair score = 0.16 (21$\times$ ratio)

The transformation occurs within the first few epochs, well before training loss plateaus.

B Extended Theoretical Results

B.1 Null Model (Corollary)

Corollary 2 (Null Model)**.** *For randomly initialized weights with no training, all pairs satisfy* $\mathbb{E}[s(i,j)] = 1/\sqrt{d} + o(1)$ *uniformly in* (i,j)*. Hungarian matching recovers correct pairs at the chance rate* $1/L$*.*

The null model rules out three alternative explanations:

1. Diagonal dominance is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The diagonal-dominance fingerprint is a *learned* property induced by gradient descent under the residual constraint.

B.2 Geometric Interpretation of the Score

The diagonal-dominance score $s(M) = |\mathrm{tr}(M)|/\|M\|_F$ has a natural geometric interpretation:

- A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$
- A matrix proportional to the identity achieves $s = \sqrt{d}$

The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score at $\Theta(\sqrt{d})$, while incorrect pairings remain at the random baseline $\Theta(1/\sqrt{d})$. This yields a signal-to-noise gap that grows linearly with hidden dimension d, making correct pairs increasingly separable in wider networks.

For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i,i) \in [1.76, 3.23]$—well above the baseline but below the theoretical maximum of $\sqrt{d}$.

C Verification Protocol Details

C.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong diagonal dominance. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\Big(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\Big) \quad (9)$$

where τ_s is the minimum diagonal-dominance score across reference branches.

C.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}} \quad (10)$$

Under Gaussian approximation, $\tau = 1.645$ yields 95% confidence; $\tau = 3.0$ yields FPR $< 0.13\%$ for high-stakes applications.

C.3 Full Protocol Steps

1. **Compatibility Check:** Verify $\text{arch}(A) = \text{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}, \{M_\ell^B\}$.
3. **Fingerprint:** Compute centered residual signatures $\{\phi_\ell^A\}, \{\phi_\ell^B\}$.
4. **Align:** Apply Hungarian matching to recover block correspondences.
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$.
6. **Calibrate:** Compute z-score $Z(A, B)$.
7. **Decide:** Return verdict:

$$\text{Verdict} = \begin{cases} \text{DESCENDANT} & \text{if } Z(A,B) > \tau_{\text{upper}} \\ \text{NON-DESCENDANT} & \text{if } Z(A,B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (11)$$

C.4 Failure Modes

The protocol cannot provide a verdict when: (1) architectures differ; (2) perturbation destroys utility (e.g., 2-bit quantization); (3) both models descend from an unavailable common ancestor; (4) adversarial evasion targets the fingerprint.

C.5 Lineage Chain and Branching Tree

To test multi-generational ancestry recovery, we construct a branching tree: root C_1, siblings C_2, C_3 fine-tuned from C_1, grandchildren $C_4 = C_2 \to$ FT, $C_5 = C_3 \to$ FT. The lineage score recovers correct ancestry: for C_4, ranking by $\mathcal{L}(C_4, \cdot)$ is parent (0.96) $>$ grandparent (0.91) $>$ uncle (0.87) $>$ cousin (0.85) $>$ independent (0.08). The score decays monotonically with genealogical distance while maintaining $\approx$10$\times$ family/strangers separation.

D Per-Architecture Details

D.1 GPT-2 Scaling

Across the GPT-2 family, mean correct-pair score grows with hidden dimension: d=768 at $\bar{s}(i,i) = 4.18$, d=1024 at 5.46, d=1280 at 6.98, d=1600 at 7.77. The scaling matches the $\Theta(\sqrt{d})$ prediction within $\leq 8\%$.

D.2 Alternative Methods Comparison

Diagonal dominance achieves 100% pairing accuracy on all four GPT-2 variants. Frobenius norm matching degrades from 67% (GPT-2) to 47% (GPT-2-large). Singular-value distance performs at chance (0–8%). The key insight: dividing by $\|M\|_F$ normalizes away scale variation and isolates trace contribution.

D.3 ResNet Factorization

For Bottleneck blocks, the naive $W_3 W_1$ achieves chance-level accuracy. The triple product $W_3 W_2 W_1$ recovers:

- ResNet-50/layer3 (n=5): 100%, AUC 1.000
- ResNet-101/layer3 (n=22): 100%, AUC 0.995
- ResNet-152/layer3 (n=35): 91%, AUC 0.964

D.4 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 descendants (noise $1{-}10\%$, pruning $20{-}80\%$, quantization $16{-}256$ levels). Results: AUROC=1.000, TPR@1%FPR=100%. Descendant scores: $\mathcal{L} \in [0.695, 1.000]$; independent baseline: $\mathcal{L}_{\max} = 0.004$.

D.5 Branching Ancestry Recovery

On a synthetic tree $C_1 \to \{C_2, C_3\}$, $C_2 \to C_4$, $C_3 \to C_5$, the lineage score ranks candidates correctly:

$$\begin{aligned} \mathcal{L}(C_4, C_2) = 0.961 \; > \; \mathcal{L}(C_4, C_1) = 0.910 \\ > \; \mathcal{L}(C_4, C_3) = 0.867 \; > \; \mathcal{L}(C_4, C_5) = 0.850 \\ \gg \; \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance while maintaining $\approx$10$\times$ family/strangers separation.

E Attack and Robustness Details

E.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \; \mathcal{L}_{\cos}(A, A + \Delta) \; + \; \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \quad (12)$$

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 9: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires $\geq$12% utility loss.

E.2 Reparameterization Invariances

Permutations: $W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^\top$ leaves $M = W_{\text{out}}W_{\text{in}}$ unchanged.

Rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$ preserves M exactly.

Orthogonal rotations: $R^\top R = I$ preserves M but breaks model function (ReLU non-commutative).

All three yield $\mathcal{L} = 1.0$ to numerical precision—these are invariances, not evasion paths.

F Practical Applications

The fingerprint enables applications beyond forensic verification.

F.1 Training Quality Assurance

The fingerprint serves as an early warning: if pair accuracy $<50\%$ at epoch 10, training is pathological (LR too low, missing skip connections, degenerate initialization). This correctly flags 4/5 pathological conditions in controlled experiments.

F.2 Zero-Knowledge Ownership Proofs

The E matrix ($M = -\varepsilon I + E$) provides a commitment scheme: owners commit to Hash$(E_k \| r_k)$ at registration, reveal challenged blocks on request. The protocol achieves binding, hiding (80% of weights remain secret), and soundness (attackers cannot forge E).

F.3 Compression Auditing

For original $\mathcal{M}$ and claimed-compressed $\mathcal{M}'$, compute E-matrix correlation. Quantization ($r > 0.97$), pruning ($r > 0.75$ at 70%), and fine-tuning ($r > 0.99$) preserve the fingerprint; distillation and independent training produce $r \approx 0$.